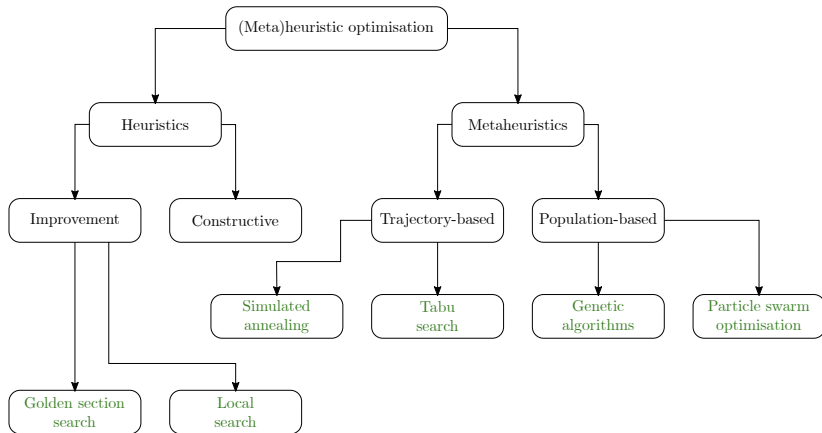


LOCAL SEARCH

Lecture 22: Presented by Jacomine Grobler



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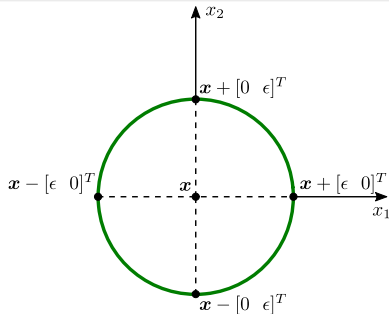


The search space $\mathbb{R}^n$ is discretised according to some regular lattice structure (usually a square grid or an hexagonal grid in the special case where $n = 2$, or a cubic grid in the special case where $n = 3$).

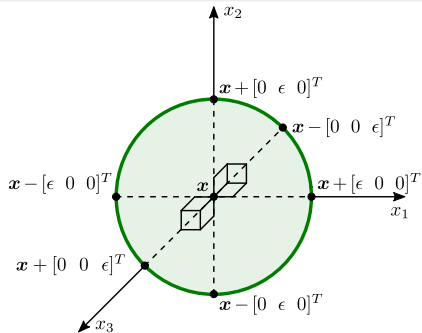
The search space $\mathbb{R}^n$ is discretised according to some regular lattice structure (usually a square grid or an hexagonal grid in the special case where $n = 2$, or a cubic grid in the special case where $n = 3$).

The neighbourhood of a candidate solution $\mathbf{x}$ represented by any lattice point is then taken as the set of all points adjacent to $\mathbf{x}$ in the lattice.

Local search

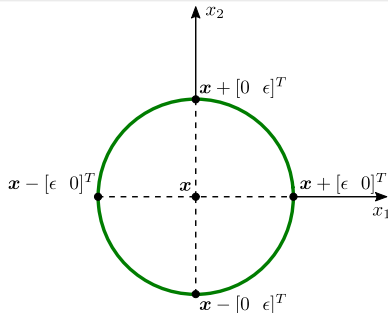


(a) Two dimensions

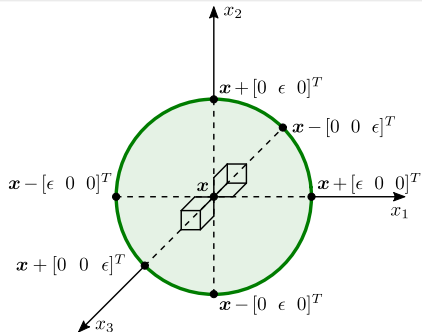


(b) Three dimensions

Local search



(a) Two dimensions

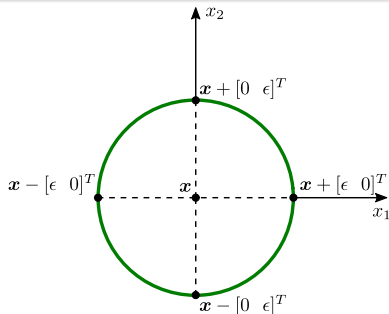


(b) Three dimensions

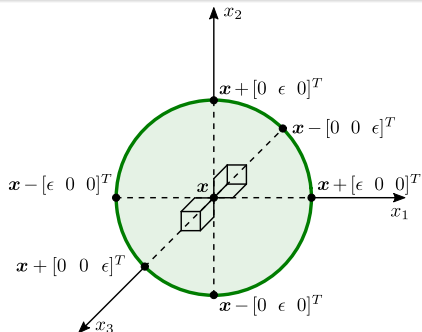
In the case of the discretisation of $\mathbb{R}^2$, the neighbourhood of the central lattice point $\mathbf{x}$ is

$$\mathcal{N}_\epsilon(\mathbf{x}) = \{\mathbf{x} - [\epsilon \ 0]^T, \mathbf{x} + [\epsilon \ 0]^T, \mathbf{x} - [0 \ \epsilon]^T, \mathbf{x} + [0 \ \epsilon]^T\}.$$

Local search



(a) Two dimensions



(b) Three dimensions

In the case of the discretisation of $\mathbb{R}^3$:

$$\mathcal{N}_\epsilon(\mathbf{x}) = \{\mathbf{x} - [\epsilon \ 0 \ 0]^T, \mathbf{x} + [\epsilon \ 0 \ 0]^T, \mathbf{x} - [0 \ \epsilon \ 0]^T, \mathbf{x} + [0 \ \epsilon \ 0]^T, \\ \mathbf{x} - [0 \ 0 \ \epsilon]^T, \mathbf{x} + [0 \ 0 \ \epsilon]^T\}.$$

In general, the neighbourhood of the central lattice point $\mathbf{x}$ is

$$\mathcal{N}_\varepsilon(\mathbf{x}) = \{\mathbf{x} \pm \varepsilon \mathbf{e}_j, j = 1, \dots, n\},$$

where $\mathbf{e}_j$ is the unit n -vector containing a one in position j , and zeros in all other positions. This neighbourhood contains $2n$ perturbations of $\mathbf{x}$.

Suppose the k -th search point uncovered during a local search is $\mathbf{x}^k$. Then the search terminates if all lattice points in the neighbourhood $\mathcal{N}_\varepsilon(\mathbf{x}^k)$ of $\mathbf{x}^k$ have objective function values that are worse than that of $\mathbf{x}^k$.

Suppose the k -th search point uncovered during a local search is $\mathbf{x}^k$. Then the search terminates if all lattice points in the neighbourhood $\mathcal{N}_\varepsilon(\mathbf{x}^k)$ of $\mathbf{x}^k$ have objective function values that are worse than that of $\mathbf{x}^k$.

Otherwise, the next search point $\mathbf{x}^{k+1}$ is chosen as that lattice point in the neighbourhood $\mathcal{N}_\varepsilon(\mathbf{x}^k)$ with the largest objective function value (in the case of maximisation) or with the smallest objective function value (in the case of minimisation).

If a better accuracy is required, the search procedure can be repeated, starting out from the previous search termination point $\mathbf{x}^\Omega$, but adopting a finer discretisation of $\mathbb{R}^n$, by performing the substitution $\varepsilon \leftarrow a\varepsilon$ for some real number $a \in (0, 1)$.

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This repetition of the local search process, each time increasing the discretisation resolution, can continue until a value of ε is reached that is smaller than some acceptable accuracy threshold value ε_{tol} .

Algorithm 4.3 (Local search heuristic)

Input: A multivariate function $f : \mathbb{R}^n \mapsto \mathbb{R}$, an initial estimate $\mathbf{x}^0$ of a local optimum of f , a grid dimension reduction factor $a \in (0, 1)$, an initial lattice dimension parameter $\varepsilon > 0$ and an accuracy threshold value $\varepsilon_{\text{tol}} > 0$

Output: An approximation $\mathbf{x}^\Omega$ of a local optimum of the function f to within an accuracy $\|\mathbf{x}^* - \mathbf{x}^\Omega\|_1 < 2\varepsilon_{\text{tol}}$

Step 1 $k \leftarrow 0$

Step 2 Generate the lattice neighbourhood $\mathbf{x}^k$ as $\mathcal{N}_\varepsilon(\mathbf{x}^k) = \{\mathbf{x} \pm \varepsilon \mathbf{e}_j, j = 1, \dots, n\}$

Step 3 $\bar{\mathbf{x}}^k \leftarrow$ element of $\mathcal{N}(\mathbf{x}^k)$ achieving the best value of f

Step 4 If $f(\bar{\mathbf{x}}^k)$ is better than $f(\mathbf{x}^k)$, then do

Step 5 $\mathbf{x}^{k+1} \leftarrow \bar{\mathbf{x}}^k, k \leftarrow k + 1$ and return to Step 3

Step 6 Else if $\varepsilon < \varepsilon_{\text{tol}}$, then output $\mathbf{x}^k$ and stop

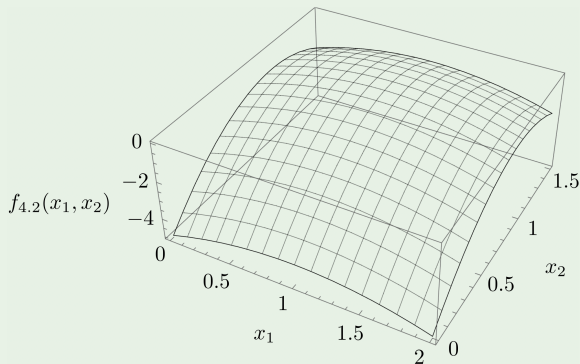
Step 7 Else $\varepsilon \leftarrow a\varepsilon$ and return to Step 2

Example 4.4 (pp. 120–123)

Apply a local search to estimate the maximum of the function

$$f_{4.2}(x_1, x_2) = -x_1^2 - 4x_2^2 + 2x_1 + 8x_2 - 5$$

starting out from the initial estimate $\mathbf{x}^0 = [0, \frac{1}{2}]^T$. Take the initial lattice cell lengths as $\varepsilon = 0.3$, halving this value whenever the search stagnates at a local maximum of $f_{4.2}$ ($a = 0.5$), and terminate the search as soon as $\varepsilon < 0.01$ (*i.e.* take $\varepsilon_{\text{tol}} = 0.01$).



Example 4.4 (pp. 120–123)

